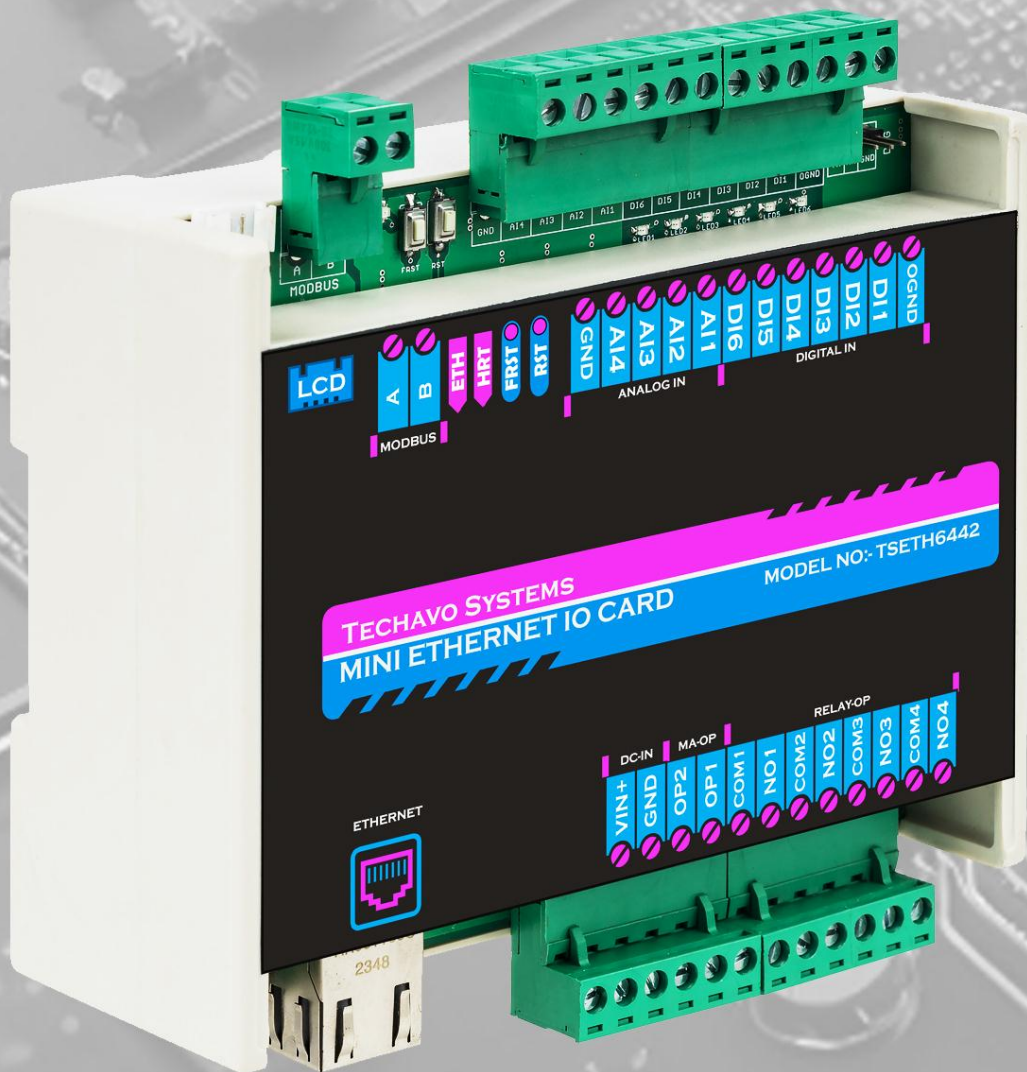


Techavo Systems

We Engineer Your Dreams

ETHERNET MODBUS RTU/IO CARD TSRTU6660



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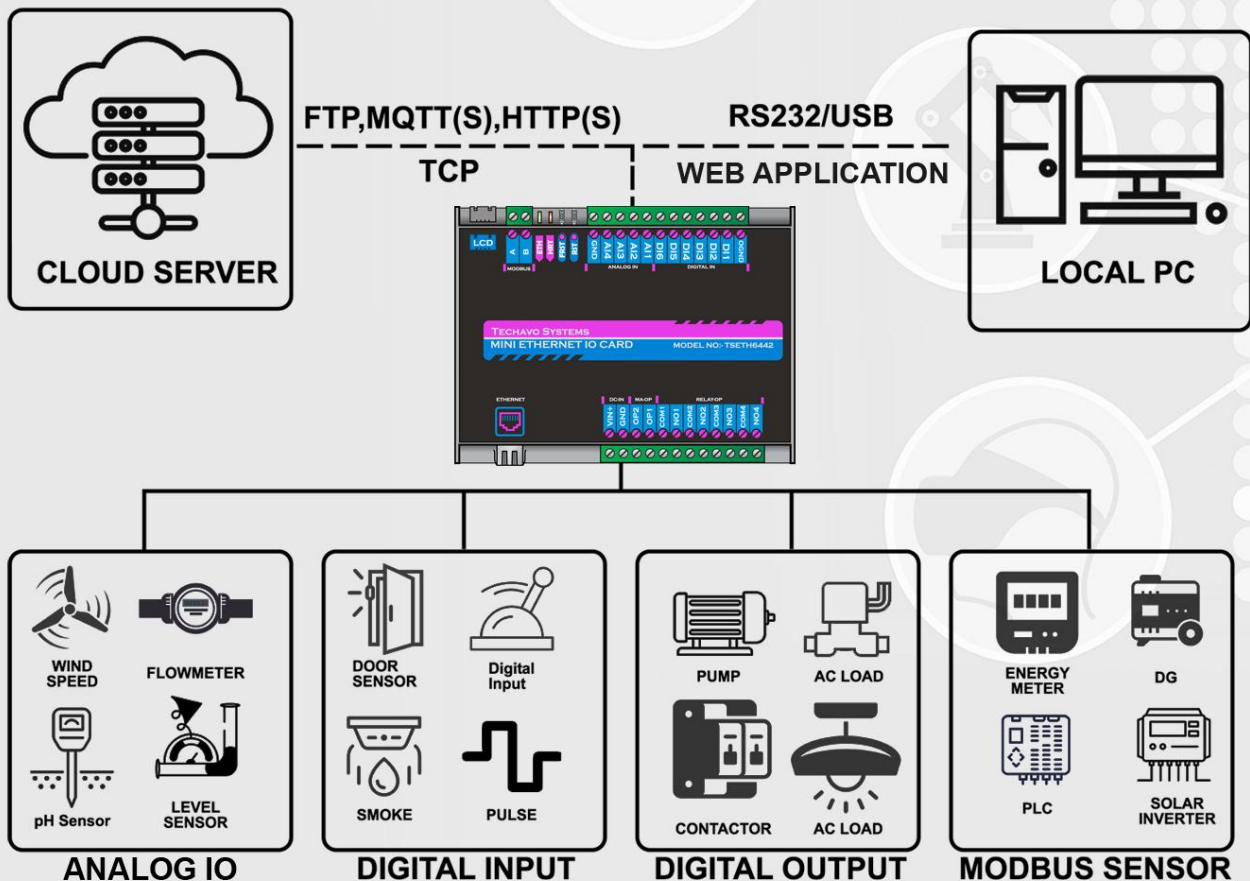
OVERVIEW

Techavo IoT data loggers are India's one of the smartest gateways manufactured in-house which can collect data from any MODBUS enabled energy meters, weather monitoring stations, variable frequency drives, string monitoring systems and all analog and digital sensor and send data to Cloud Server via several protocols like FTP, HTTP(s), MQTT(s), TCP in JSON/CSV format along with SMS and CALL notification. The device can be remotely configured using Web Application, Desktop Application, TCP, MQTT, FTP. It ensures zero data packet loss with its EDGE storage capability during Network loss.

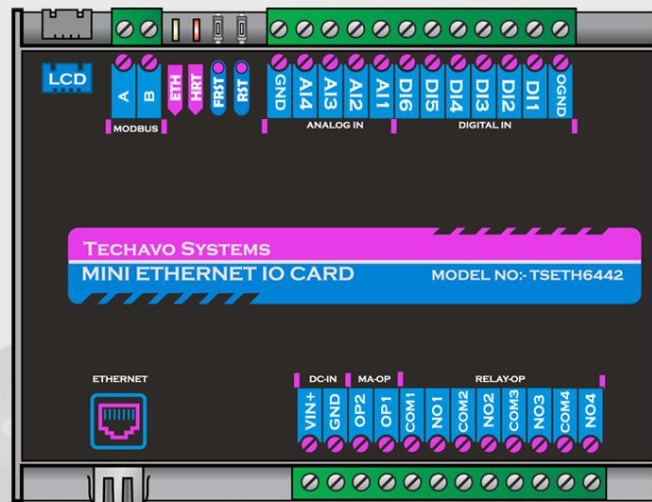
KEY FEATURES

- ➔ Comes with RJ45 Connectivity.
- ➔ Configurable upto 30 Modbus Slaves with all possible data formats, byte order, scale factor, baud rate, data bit, parity.
- ➔ Multi-Communication Protocol support (MODBUS RTU Master, Modbus RTU Transparent, Modbus RTU Slave, Modbus RTU over TCP as client & Master, Modbus TCP, HTTP(S), MQTT(S), DHCP, TCP, UDP, Ethernet.
- ➔ Data Packet is in configurable JSON/CSV format.
- ➔ Several Interlocking Logics based on AI, DI, DO available that can replace PLC.
- ➔ Upto Eight Sockets Available which can be individually configured.
- ➔ Offline Data Storage upto 64MB.
- ➔ OTA device configuration.
- ➔ Integrates with all Analog (0-10V, 4-20mA) and Digital (Dry, Wet, Pulse) Input Sensor.
- ➔ Pendrive/Sd Card Additional Storage (Optional).

APPLICATION DIAGRAM



SPECIFICATIONS



SYSTEM

Controller	Arm Processor Cortex M1 512KB Flash 32Kb SRAM
Flash	64 MB Offline Data Log memory
EEPROM	512 Kb
Real Time Clock	External with Battery Backup
Watchdog	Internal

IO PORTS

Power Supply	12~24 Volt with Reverse Protection
IO Port	4 Channel 4~20mA/0-10V Analog Input 12bit 1000SPS Non Isolated 3 Channel Relay Output with 5 Amp Max Rating 6 Channel Isolated Digital Input 2 Channel 0~10V/4~20mA Output
Led Indication	GSM,Heart Beat,Server
Display	4 Pin LCD Connector
RS485	Isolated 1 Channel RS485 Port
Serial Port	TTL Pin for PC configuration

Ethernet 10/100 MBPS

Connector Type	RJ45
Isolation	1500VRMS

Communication

Protocols	Modbus RTU Polling, Trasnparent Modbus RTU over TCP/UDP as client & Server, Trasnparent Modbus TCP/UDP,
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Export

Protocols	TCP, HTTP (s), MQTT(s), FTP, DNS, DHCP.
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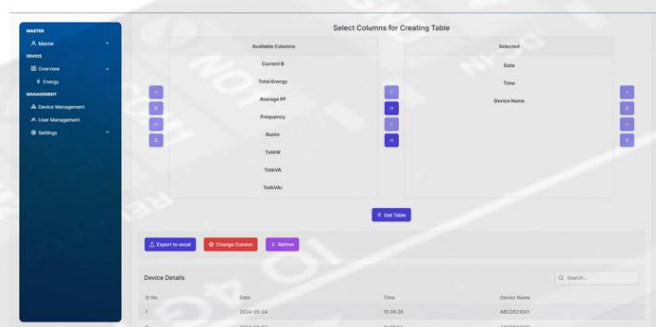
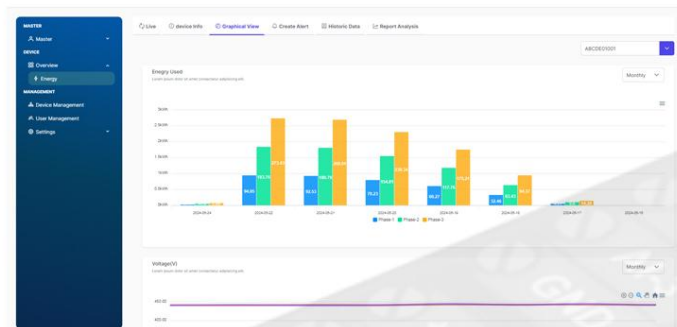
Mechanical

Casing	ABS Enclosure
Protection	IP20 (Indoor)
Dimensions (W x L x H)	70mm X 130mm x 60mm
Installation	DIN Rail Mount

Environmental

Casing	ABS Enclosure
Protection	IP20 (Indoor)
Dimensions (W x L x H)	70mm X 130mm x 60mm
Installation	DIN Rail Mount

CLOUD SOFTWARE



Sl No.	Start Date	Energy Used	Electricity Bill
1	2024-01-01 12:31:20	0.00 kWh	\$0.00
2	2024-01-17 23:58:16	28.79 kWh	\$110.3600
3	2024-01-18 23:58:58	199.26 kWh	\$761.0400
4	2024-01-19 23:58:58	262.24 kWh	\$1012.9600
5	2024-01-20 23:58:58	464.36 kWh	\$1817.0000
6	2024-01-21 11:20:05	542.03 kWh	\$2168.0000
7	2024-01-22 11:47:24	575.34 kWh	\$2254.8800
8	2024-01-23 11:40:36	147.00 kWh	\$568.0000

Key Features

Device Management & Summary

Devices can be added and deleted by admin. It will give us a summary of all the devices installed in a particular site along with geolocation and live status. This helps to find out faulty or healthy devices in a particular zone.

Live Data Monitoring

Get a holistic view via a powerful GUI that helps centralise various parameters of the device. Benchmark building performance across multiple locations, sites or even countries with this centralised dashboard, anytime and anywhere.

Graphical Representation

This will give us a graphical time graph of all the parameters. Time can be selected as daily, monthly, yearly or customized. The graph can be represented in several formats.

Historical Data

Reporting templates can be developed easily through a WYSIWYG editor, parameterised and customisable to help you filter the data to get only what you need. Create reports in every available format for all use cases, and easily automate and schedule the creation and delivery of reports to end users or customers via email or URL links.

Alarm Generation

Set multi-condition alerts via powerful, flexible and intuitive “if-then-else” conditions, turn On/Off devices automatically based on pre-set threshold parameters or warning sirens. Trigger notifications to multiple parties instantaneously via emails and/or SMS. Log these alerts so that you may review all historical data to provide an audit trail for ease of security and compliance.

User Management

This has a complete hierarchy of users and super admin can create multiple users with several selectable privileges.

Solutions We Provide:

1. Embedded Systems Development Services
2. Hardware Development for Embedded Systems
3. Firmware Development For Embedded Systems
4. Prototyping
5. Product Development
6. IoT Application Development
7. Home Automation
8. Remote Monitoring

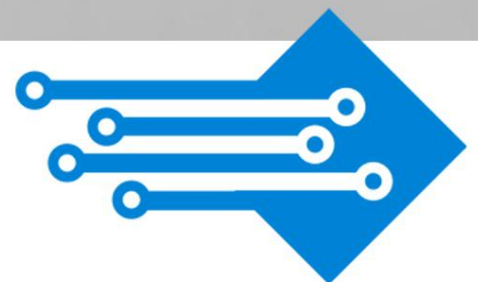
Our Range of Products:

1. Remote Terminal Unit
2. Cloud-Based Ultra-Low Power Data Logger
3. IoT Gateway
4. Cloud-Based Monitoring Solutions/Dashboarding
5. IO Cards
6. Customized Industrial Automation Units

About Us

Techavo Systems is your one-stop solution for all your IoT integration needs.

We specialize in creating seamless, cutting-edge solutions that transform your lifestyle. Our team is constantly pushing the boundaries to challenge the norm and ensure consistency and complacency. Whether you need home automation, embedded systems, or remote monitoring solutions, we have the expertise to deliver. From custom product development to IoT development and application, we've got you covered.



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